

# Software Requirements Specification

Version 1.0

<<Annotated Version>>

26-03-2026

College Education Management System

## Team Members:

Khaled Yasser

Ziad Hamed

Mohamed Osama

Noura Mostafa

Esraa Yahia

## Table of Contents

|                                           |    |
|-------------------------------------------|----|
| Table of Contents .....                   | ii |
| 1.0. Introduction .....                   | 2  |
| 1.1. Purpose .....                        | 2  |
| 1.2. Scope of Project .....               | 2  |
| 2.0. General Description .....            | 2  |
| 2.1 Product Perspective .....             | 2  |
| 2.2 Product Functions .....               | 2  |
| 2.3 User Characteristics .....            | 2  |
| 2.4 General Constraints .....             | 3  |
| 3.0. Requirements Specification .....     | 3  |
| 3.1 External Interface Requirements ..... | 3  |
| 3.1.1 User Interface (UI) .....           | 3  |
| 3.1.2 Hardware Interfaces .....           | 3  |
| 3.1.3 Software Interfaces .....           | 3  |
| 3.1.4 Communication Interfaces .....      | 3  |
| 3.2 Functional Requirements .....         | 3  |
| 3.3 Non-Functional Requirements .....     | 4  |

## 1.0. Introduction

### 1.1. Purpose

This document defines the functional and non-functional requirements for the College Education Management System which streamlines academic operations by managing students, doctors, courses and departments.

### 1.2. Scope of Project

The College Education Management System is a web-based application designed to streamline and automate core academic operations at the College.

The system provides centralized management for:

- . Student and Doctor information, including personal details and enrollment status.
- . Courses and enrollment, enabling students to register for courses and view course materials.

- . Academic report generation, supporting administrative decision-making with detailed insights on doctors, students and courses.
- . Grade recording and GPA calculation.

## **2.0.General Description**

### ***2.1 Product Perspective***

- . The College Education Management System is a standalone web-based application designed to manage student records, courses, enrollment, and grading.
- . It uses a centralized database to store and retrieve information and can send notifications.
- . The system serves admins, doctors, and students, each with specific access and functions.

### ***2.2 Product Functions***

- . Manage student and doctor information (add, update, delete).
- . Create and manage courses.
- . Enroll and unenroll students in courses.
- . Assign and unassign doctors to/from courses.
- . Record and calculate grades.
- . Generate reports for students and doctors.

### **2.3 User Classes and Characteristics:**

#### **Admin:**

- . Full access to all system features, including adding, updating, and deleting students ,courses and doctors.
- . Requires basic to intermediate computer skills.

#### **Doctor:**

- . Can upload materials, assignments, quizzes and generate student reports.

#### **Student:**

- . Can enroll in courses, view schedule, grades, GPA, access materials, and submit assignments and quizzes.

## ***2.4 General Constraints***

- . The system is a web-based application.
- . It must support multiple concurrent users without performance degradation.

## ***3. Requirements Specification***

### ***3.1 External Interface Requirements***

#### ***3.1.1 User Interface (UI)***

- . The system shall provide a simple and user-friendly interface.
- . Admins shall be able to easily navigate between modules.
- . Confirmation messages shall be shown for important actions (add, delete, update).

#### ***3.1.2 Hardware Interfaces***

##### ***Client Side:***

- . Device: Computer or laptop.
- . Modern web browser (Chrome, Edge, Firefox).

##### ***Server Side:***

- . RAM: Minimum 4 GB (8 GB recommended).
- . Processor: Intel Core i3 or higher.
- . Storage: Minimum 500 MB free space.
- . Keyboard and mouse for interaction.

#### ***3.1.3 Software Interfaces***

- . Operating System: Windows 10 or later.
- . Programming Language: C# (.NET Framework / .NET Core).
- . Database: SQL Server.
- . IDE: Visual Studio.
- . .NET Runtime installed.

#### ***3.1.4 Communication Interfaces***

- . Requires a stable internet connection.
- . Database communication must be secure.
- . Data transfer between components shall be reliable and consistent.

### ***3.2 Functional Requirements***

#### ***Admin Functions:***

- . Login and logout.
- . Add, update, and delete students.
- . Add, update, and delete doctors.
- . Create and manage courses.
- . Assign and unassign doctors to courses.
- . Generate reports for students and doctors.
- . Send notifications to students and doctors.

#### **Doctor Functions:**

- . Login and logout.
- . Upload course materials.
- . Create assignments and quizzes.
- . Generate reports about students and send to admin.

#### **Student Functions:**

- . Login and logout.
- . Enroll and unenroll in courses.
- . View enrolled courses and course materials.
- . View schedule, grades and GPA.
- . Submit assignments and quizzes.

### ***3.3 Non-Functional Requirements***

- . **Performance:** The system must support multiple concurrent users without slowdown.
- . **Reliability:** Data must be stored securely and backed up regularly.
- . **Usability:** The interface should be user-friendly for admins, doctors, and students.
- . **Security:** User authentication is required; sensitive data must be protected.
- . **Maintainability:** The system should allow easy updates and bug fixes.
- . **Platform:** The system will run as a web application accessible via modern browsers.